Glossary

DHCP: Domain Host Control Protocol lets each computer that connects to the network get a temporary IP address. It is used in scenarios where there are more computers than available IP addresses.

MIC: Message integrity check. A one-way cryptographic hash function used in WPA instead of CRC32.

Repeater mode: In this method, an AP is an extension of remote access point. Repeater Mode is used to extend the signal range of a remote AP or wireless router. The MAC address of the remote AP is required to set an AP into repeater mode.

RF: Radio Frequency. Specifies the frequency at which the signal is being broadcast.

SNR: Signal-to-Noise Ratio is the ratio of the amplitude of desired signal to that of noise at a specific time. It is expressed in decibels.

SSID: Secure Set Identification is a name given to the Access Point. When the client PC scans for an available network, the SSID will be displayed in the 'available network' box, and the user can click on it to connect.

TKIP: Temporary Key Integrity Protocol ensures the key changes at every specified interval, and also performs a hash to generate a new key in WPA.

WEP: Wired Equivalent Privacy is a data encryption method to provide security to the client systems connecting to it. It is not very safe because of the insufficient length of the network key; besides, there's no method to distribute it amongst the clients.

Wireless bridge: In this mode, two APs can be bridged with each other for communication. It is used when the networks of two buildings or offices need to be connected wirelessly.

WPA: Wi-Fi Protected Access is an improvement over WEP, as it utilises the Temporary Key Integrity protocol (TKIP). It will be in use until the 802.11i standard is properly ratified.

select 'Allow me to connect to the selected network' and press 'connect.' The PC is now on the wireless network.

Creating a Mobile Workforce

Wireless PCMCIA cards (PC cards) let laptop-wielding executives connect to the network as they walk into office.

But then, they may not need a PC card, as most of the new laptop models are already Wi-Fi-enabled. If your company uses such laptops, configure the cards to connect to the AP and you are done.

In the tests, we had six PCMCIA cards—three each for 802.11g and 802.11b.

PC CARDS

Performance

Performance-wise, the PCMCIA cards did not put up a great show as compared to PCI cards. A factor that definitely affected performance was the presence of an

NetGear WG511

AirStation WLI-CB-G54A

external antenna in PCI cards, which is absent in the PCMCIAs, especially at Location 3. In the 802.11g category, the NetGear WG511 outperformed the others. Similarly, the Buffalo WLI-CB-G54A easily came out tops in the 802.11b category. In the compatibility test, Compex's WL54G failed badly at Location 3.

Features

By default, PC cards do not have an external antenna, but Compex's WL54G and Buffalo's WLI-CB-G54A did provide ports to connect them. Only the D-Link in the 802.11g category and Linksys' WPC11 in the 802.11b category had WPA.

Price

Here, the NetGear and Buffalo took a big lead in the performance tests, and so, despite the low price of the D-Link and the Compex, the NetGear and Buffalo devices took the crown in this category.

Enabling Wireless

Just like PCI cards, PC cards too are simple and straightforward to install. Just plug in the PC card into the slot in the notebook, insert the driver CD in the CD-ROM and let the installation complete. The difference here is that you can